Lenguajes Formales, Autómatas y Computabilidad

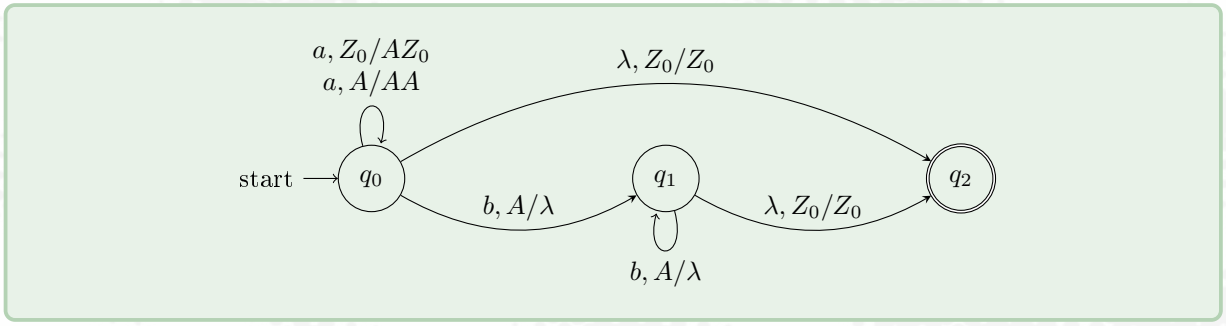
Practica 5: Automatas de Pila

2 de noviembre de 2026

Ejercicio 1. Para cada uno de los siguientes lenguajes, construir un automata de pila que los acepte. Hacer una version determinística en los casos en que sea posible.

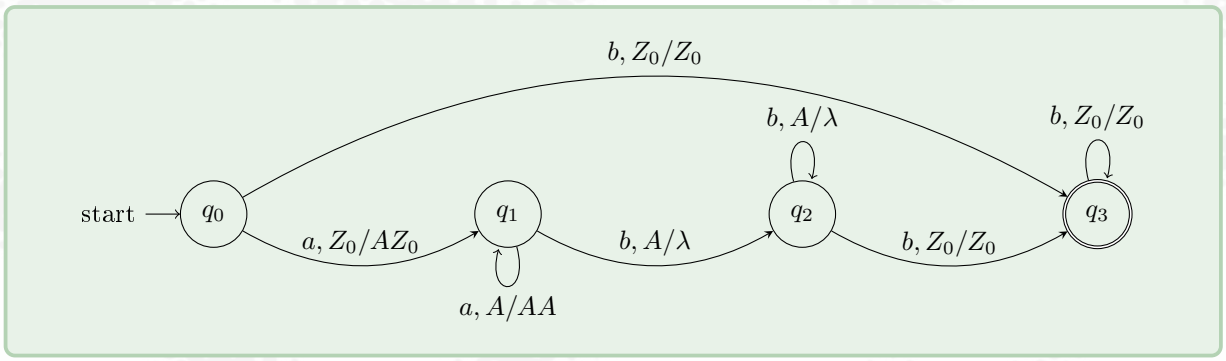
a) $\{a^n b^n\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



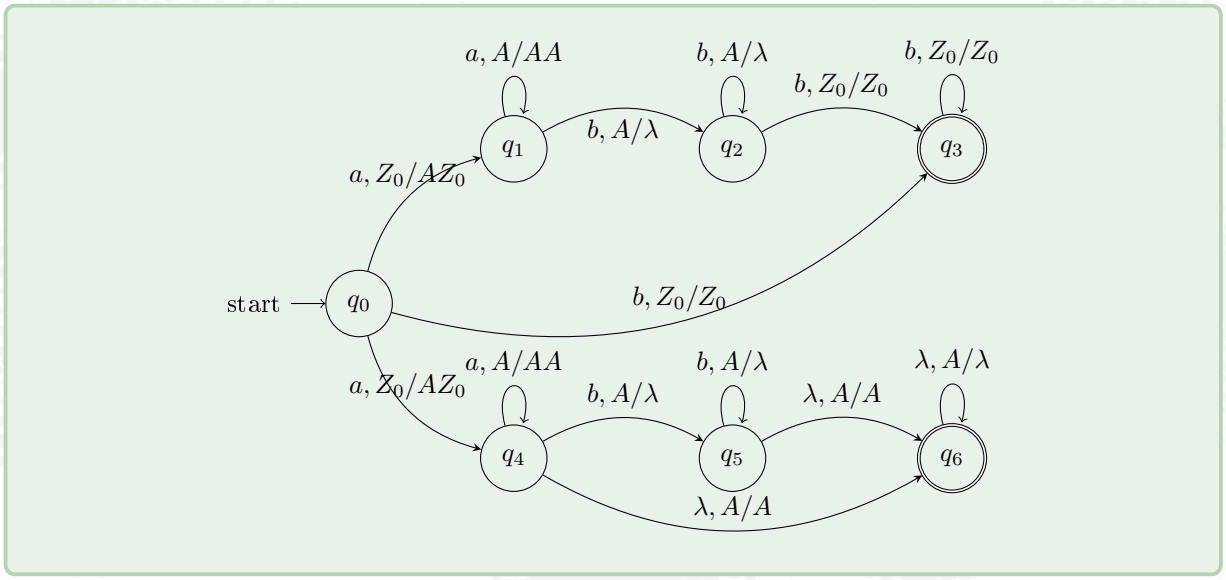
b) $\{a^n b^m \mid m > n\}$

Sol: $\langle \{q_0, q_1, q_2, q_3\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_3\} \rangle$



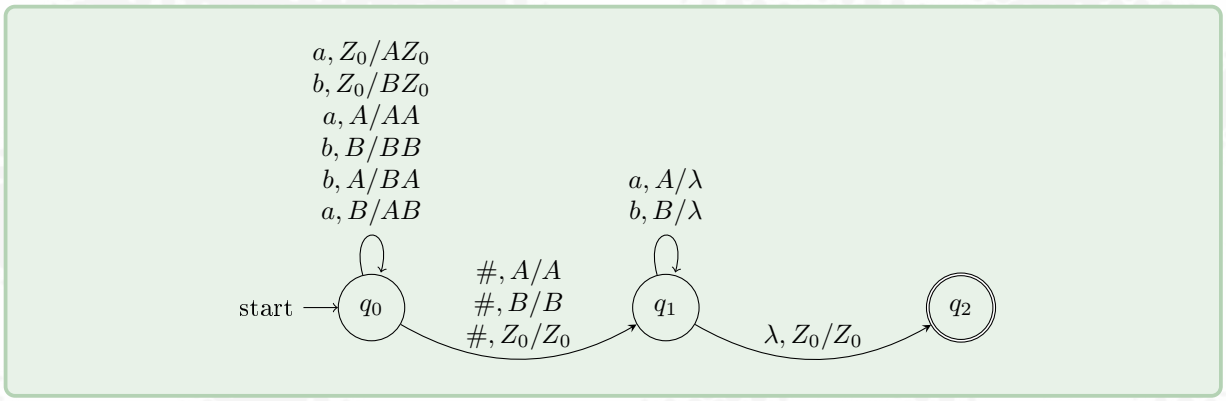
c) $\{a^n b^m \mid m \neq n\}$

Sol: $\langle \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_3, q_6\} \rangle$



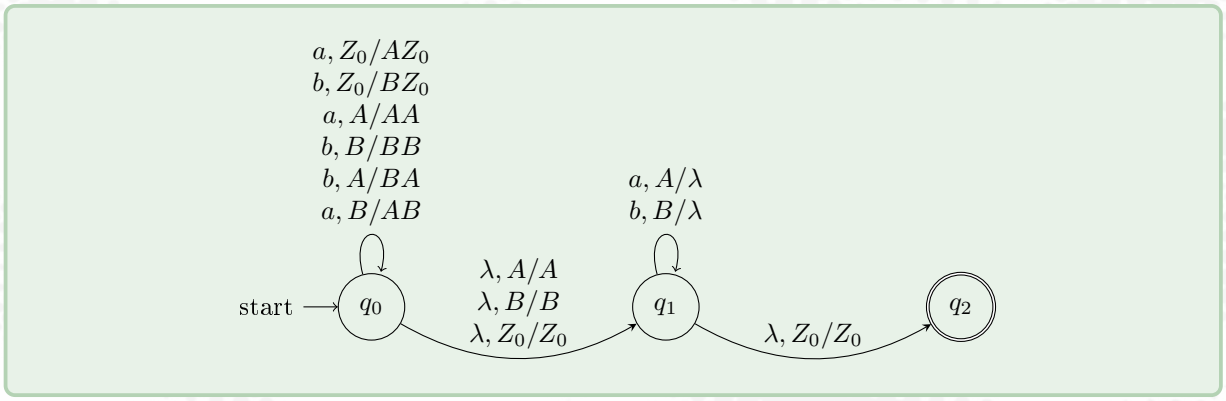
d) $\{\omega\#\omega^r \mid \omega \in \{a,b\}^*\}$

Sol: $\langle\{q_0,q_1,q_2\},\{a,\#,b\},\{A,B,Z_0\},\delta,q_0,Z_0,\{q_2\}\rangle$



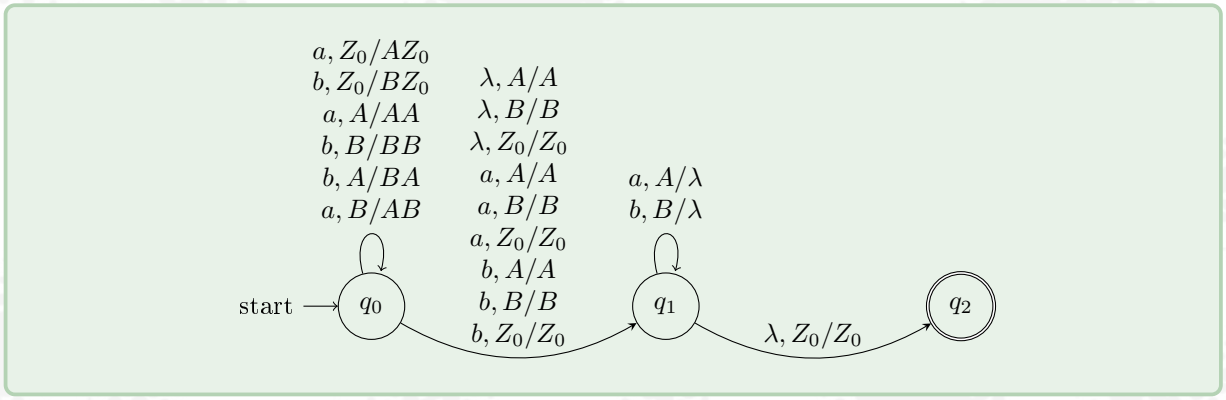
e) $\{\omega\omega^r \mid \omega \in \{a,b\}^*\}$

Sol: $\langle\{q_0,q_1,q_2\},\{a,b\},\{A,B,Z_0\},\delta,q_0,Z_0,\{q_2\}\rangle$



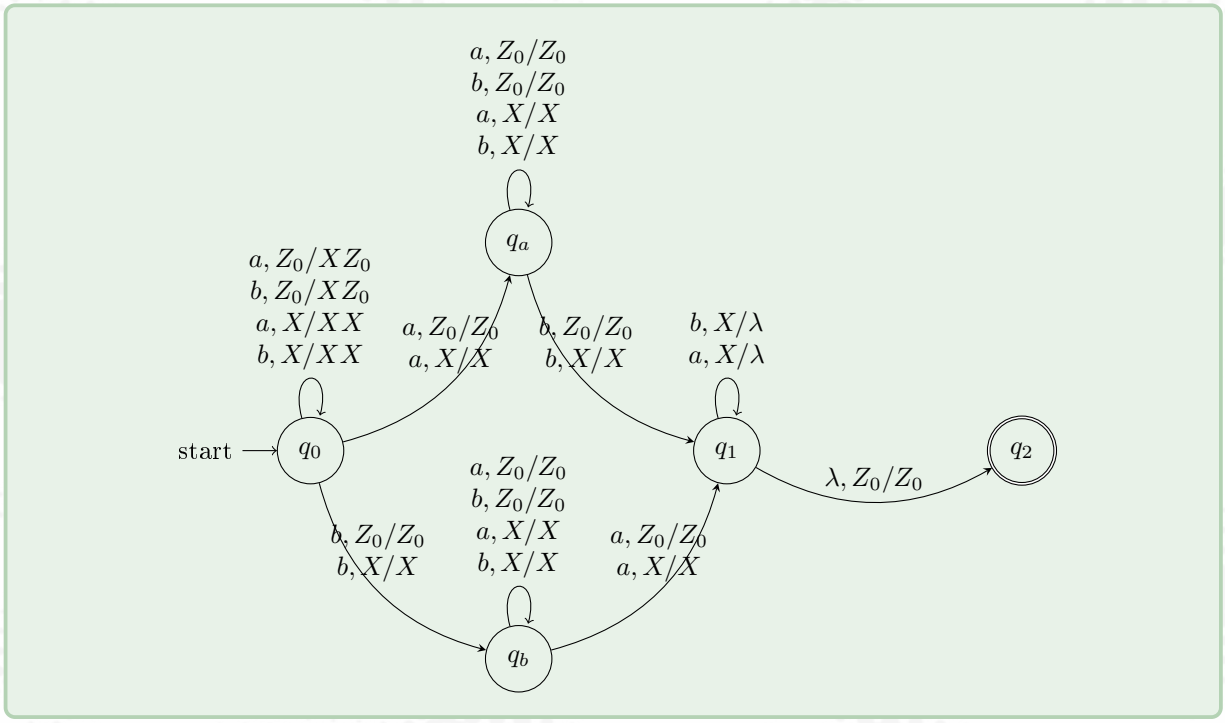
f) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega = \omega^r\}$

Sol: $\langle\{q_0,q_1,q_2\},\{a,b\},\{A,B,Z_0\},\delta,q_0,Z_0,\{q_2\}\rangle$



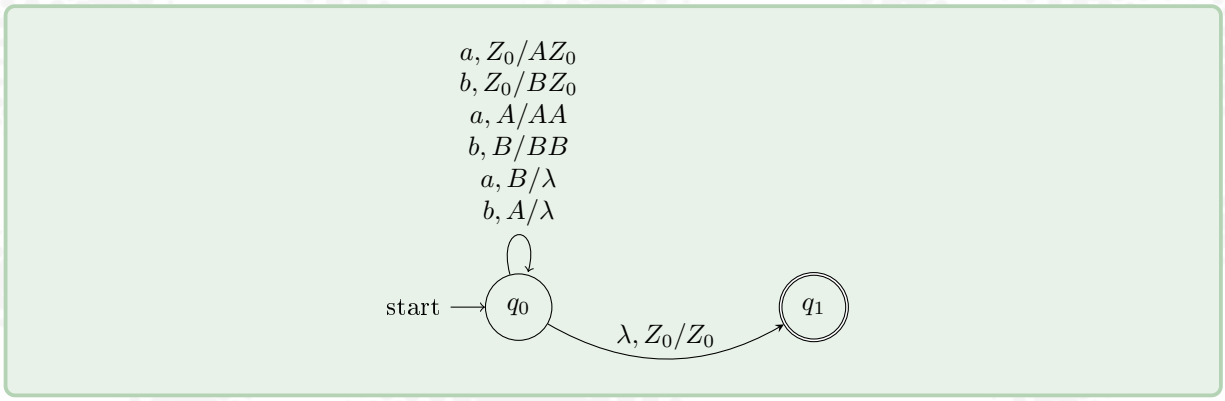
g) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega \neq \omega^r\}$

Sol: $\langle\{q_0,q_a,q_b,q_1,q_2\},\{a,b\},\{X,Z_0\},\delta,q_0,Z_0,\{q_2\}\rangle$



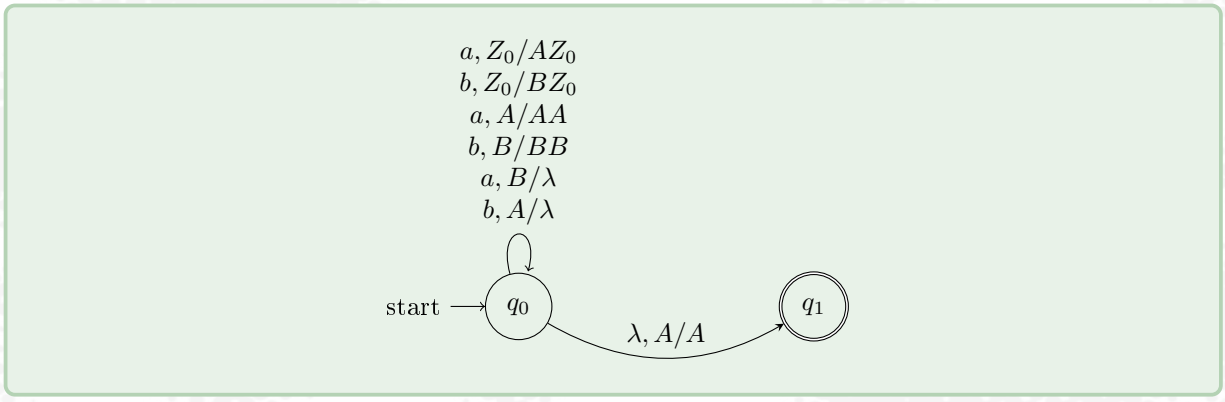
h) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega_a = \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a,b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



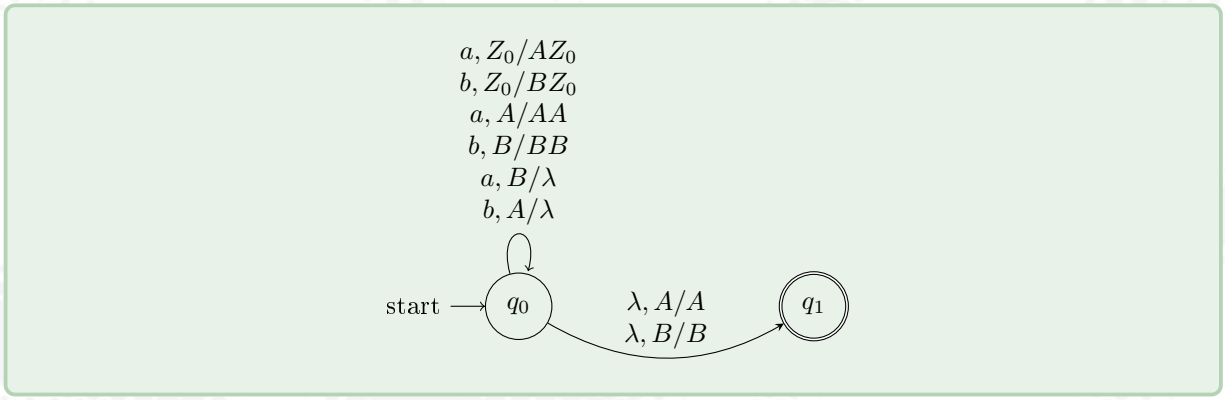
i) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega_a > \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a,b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



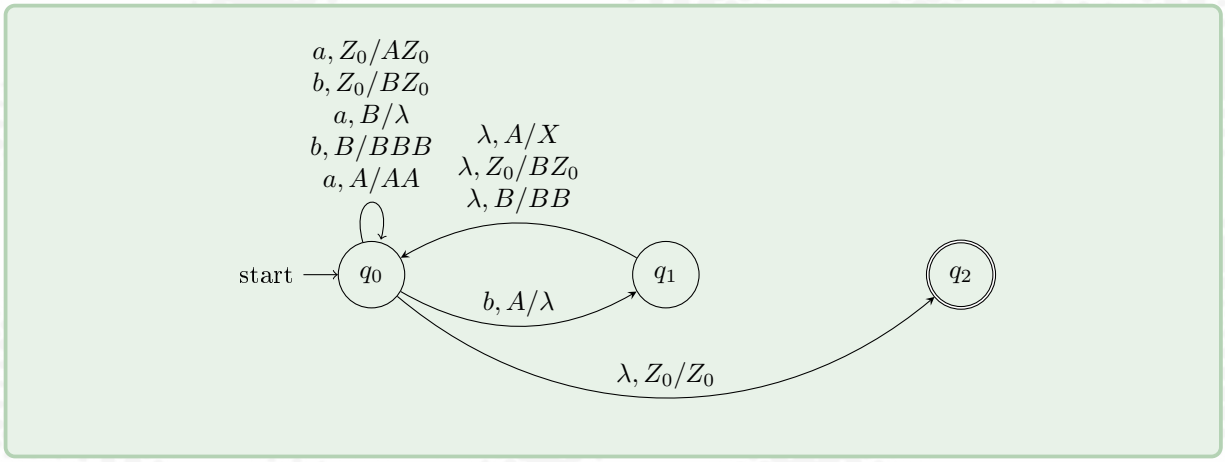
j) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega_a \neq \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a,b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



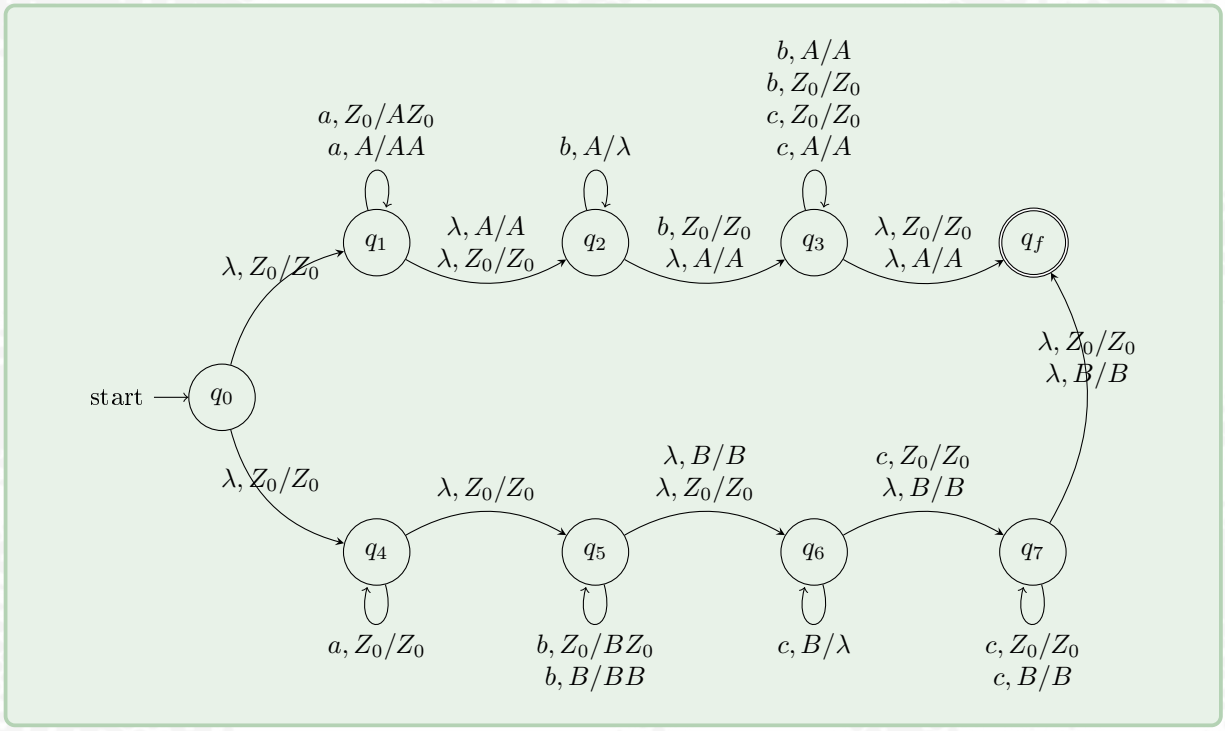
k) $\{\omega \mid \omega \in \{a,b\}^* \wedge \omega_a = 2\omega_b\}$

Sol: $\langle\{q_0, q_1, q_2\}, \{a,b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_2\}\rangle$



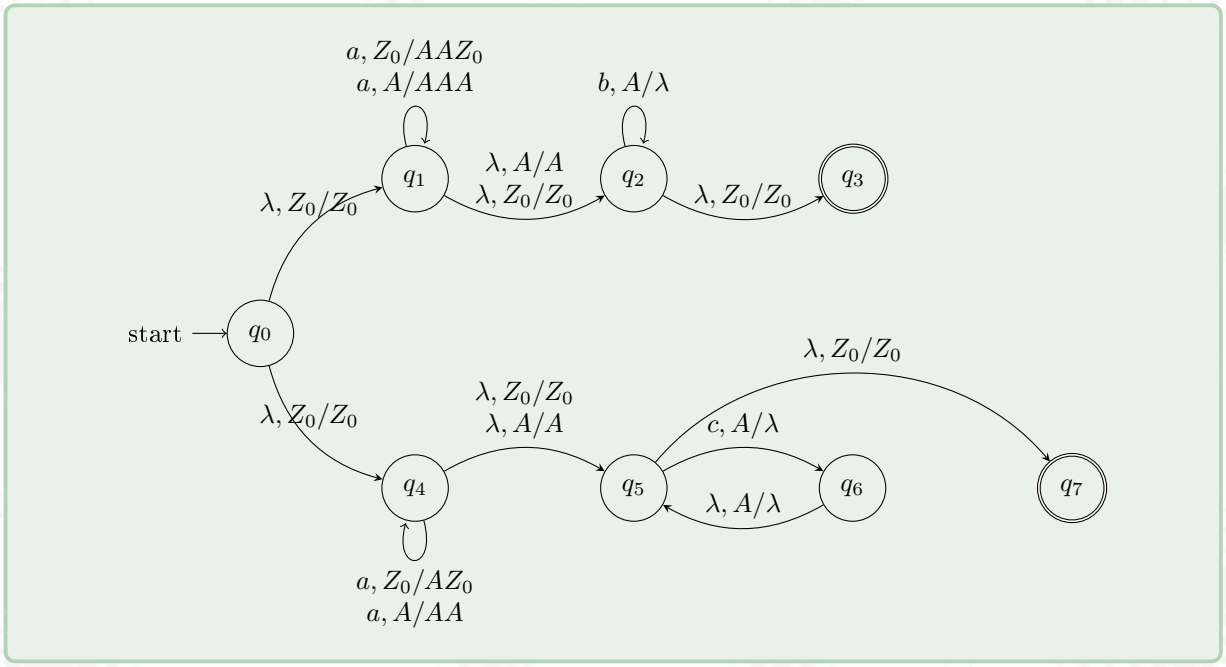
l) $\{a^n b^m c^k \mid n \neq m \vee m \neq k\}$

Sol: $\langle\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_f\}, \{a,b,c\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_f\}\rangle$



m) $\{a^n b^m \mid m = 2n\} \cup \{a^n c^m \mid n = 2m\}$

Sol: $\langle\{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}, \{a,b,c\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_7\}\rangle$



n) Cadenas de paréntesis balanceados.

Sol: $\langle \{q_0, q_1\}, \{(,)\}, \{X, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$

